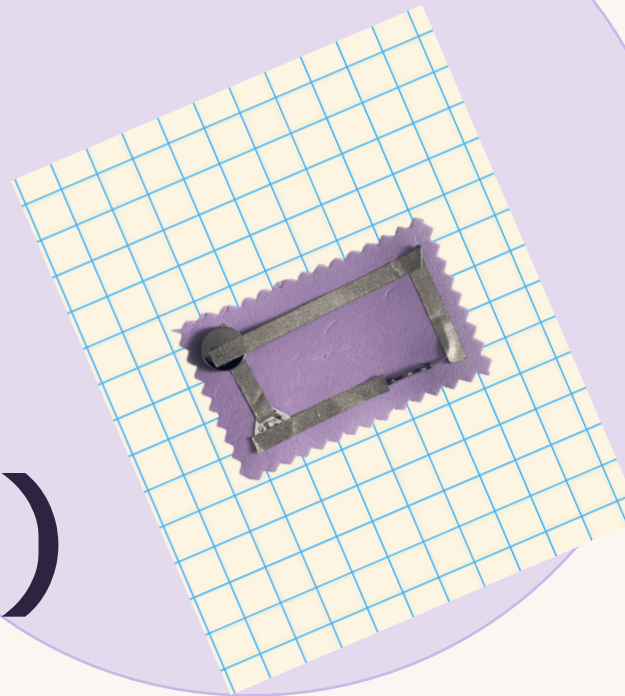
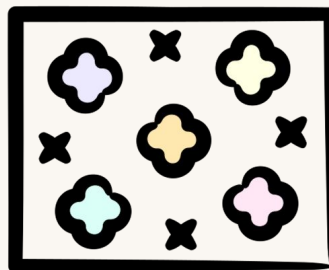


ITP Camp 2026

Soft Circuit Boards (SCBs)



handmade circuits on craft foam

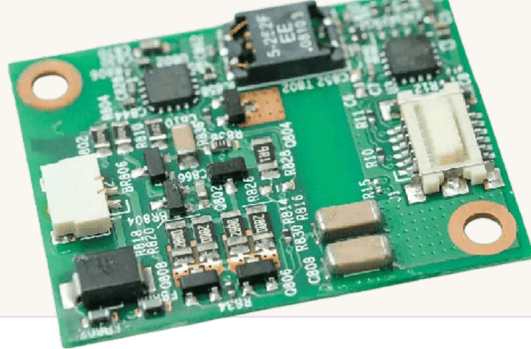


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SCB ✨

context

what is a PCB?



Printed Circuit Board

a flat board that connects electronic components using conductive pathways called traces.

usually made of fiberglass with thin copper etched onto the surface.

anatomy of a PCB

substrate

the board itself
(fiberglass)

copper layer

thin conductive sheet

traces

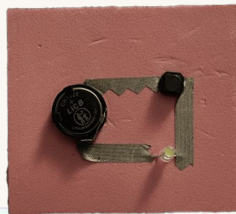
paths that carry
electricity

pads

where components attach

what we're making

Soft Circuit Board



a circuit board made by hand

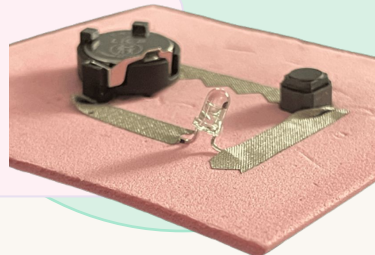
copper tape traces on craft foam – no etching, no soldering, no factory

circuits you can see, touch, and repair

every trace is visible. every connection is physical. nothing is hidden.

same logic as a PCB, built with craft materials

battery, traces, components – it all works the same way.



what do traces do?

think of traces like roads –

electricity travels from battery → component → back to battery,
just like cars travel on roads.

on a real PCB

copper trace

drilled pad

solder

FR4 fiberglass

→

→

→

→

on our SCB

copper tape

poked hole

pressure + tape

craft foam

what you have

materials



craft foam

your board



copper tape

your traces



LEDs

the light



CR2032 battery

power source



**scissors + pen
(optional)**

to cut & trace



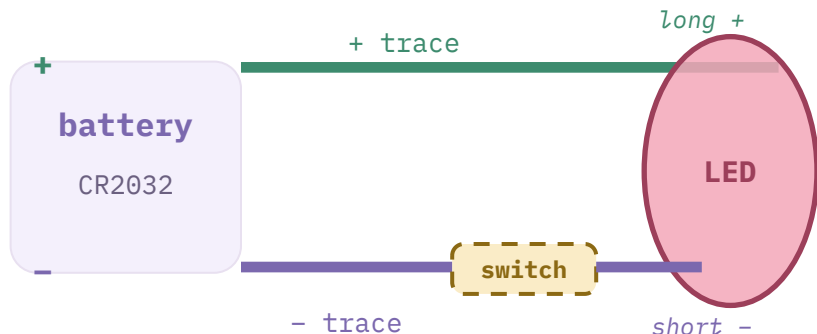
**button
(optional)**

a switch



the circuit

a simple LED circuit



key rules

- circuit must be a complete loop
- long LED leg → battery +
- short LED leg → battery -
- + and - must not touch
- switch opens and closes the loop

step by step

how to build your SCB

1 sketch first

lightly draw your trace paths on the foam. mark where the LED and battery go.

2 lay copper tape

follow your sketch. press firmly. fold at corners – don't cut.

3 leave the gap

leave a ~1cm gap in your – trace. this is your switch.

4 poke in the LED

push legs through foam where traces meet. long leg on + side.

5 add the battery

slide coin cell between trace ends. + face up on + side.

6 test it!

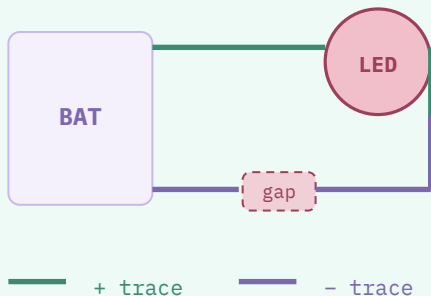
press battery down. touch the gap with your finger. light?

let's craft it! ✨

beginner

touch switch LED

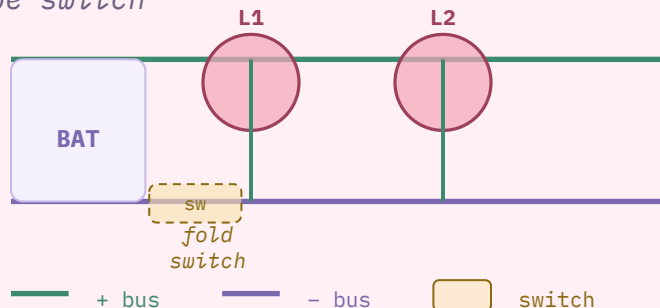
LED + coin cell + copper tape + switch



experienced

parallel LEDs + fold switch

two LEDs, shared + and - rails, folded tape switch



share out ✨

Show off your circuit!

what would you build next?

**what was the trickiest
part?**

what would you change?

thanks for crafting!!! ✨

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